

Drone Delivery Route Optimization System

A visual, easy-to-explain project brief covering the project idea, system behavior, algorithms/data structures, and main benefits.



1 Project in Simple Words

Our project is a **Drone Delivery Route Optimization System** that helps delivery drones find the most efficient routes to deliver packages in a city. The system automatically decides which delivery to do first, which route to take, and how to divide tasks among multiple drones while considering distance, travel time, and battery (energy) usage.

"It's like a smart planner for delivery drones."



Project idea: Build a software simulation that helps delivery drones choose efficient routes, prioritize deliveries, manage battery/energy usage, and divide delivery work among multiple drones.

1

Project in Very Simple Words

Imagine a delivery company has 3 drones and 10 packages that need to be delivered to different locations in a city.

The company has to answer questions like:

- Which drone should deliver which package?
- Which package should be delivered first?
- What is the shortest or fastest route?
- What if the shortest route uses too much battery?
- How can several drones divide the work efficiently?
- Can the deliveries be completed without wasting time and energy?

Our Drone Delivery Route Optimization System will make these decisions automatically using algorithms and data structures.

In simple words, the system acts like a smart planner for delivery drones.

Simple Example

Three drones

Drone 1 - 90% battery
Drone 2 - 60% battery
Drone 3 - 75% battery

Five packages

Package A - Location 2 - Normal
Package B - Location 7 - Urgent
Package C - Location 5 - Normal
Package D - Location 9 - High
Package E - Location 4 - Normal

The system will decide:

- 1 Which delivery should be handled first.
- 2 Which drone is the most suitable for that delivery.
- 3 Which route that drone should take.
- 4 Whether the drone has enough battery for the route.
- 5 How the remaining deliveries should be distributed among the other drones.

2 What the System Will Do

2 What the System Will Do



Create a Simulated City Map

Represent the city as a graph where locations are nodes and paths are edges with distance, time, and energy information.



Receive Delivery Requests

Take multiple delivery tasks with different destinations and priorities.



Find the Best Route

Use algorithms like Dijkstra or A* to find the shortest and fastest route.



Consider Battery/Energy Usage

Choose routes that are energy-efficient, not just the shortest distance.



Prioritize Deliveries

Use a greedy approach and priority queue to deliver urgent packages first (e.g., medicine, food).



Divide Work Between Multiple Drones

Assign deliveries to different drones efficiently so all packages are delivered on time.



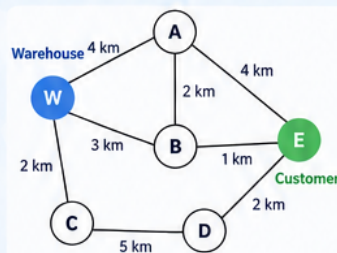
Show the Selected Route

Display the chosen route, drone details, distance, energy usage, and delivery status.

How It Works (Flow)



Example: Finding the Best Route



Possible Routes (W → E)

Route 1: W → A → E = 8 km
Route 2: W → B → E = 4 km ✓ (Shortest)
Route 3: W → C → D → E = 9 km

The system will choose **Route 2** because it is the shortest and most efficient.

Distance vs Energy Example

Route A

Distance: 5 km
Energy: 40%



Shorter, but uses more energy

Route B

Distance: 6 km
Energy: 25%



A bit longer, but more energy efficient

The workflow above shows how the project moves from delivery requests to priority handling, drone selection, route optimization, and finally delivery execution.

The system combines several smaller algorithmic decisions instead of treating delivery as only a shortest-path problem.

At a glance

City model	Weighted graph with locations as nodes and paths as edges
Routing	Dijkstra / A* to find efficient routes
Energy	Compare distance with estimated battery usage
Priority	Urgent deliveries move ahead of normal deliveries
Multiple drones	Distribute tasks across available drones
Output	Show selected drone, route, distance, energy and priority

2

What the System Will Do - Detailed

2.1 Create a Simulated City Map

The system will represent a fictional city as a Graph.

Locations such as the warehouse, customer areas, charging points, or delivery points will be represented as nodes/vertices. The possible paths between locations will be represented as edges.

Every edge can store information such as distance, travel time, and energy usage.

```
Warehouse
|
A ---- B
| |
C ---- D
|
Customer
```

This allows the system to compare different possible routes between the warehouse and a delivery destination.

2.2 Receive Delivery Requests

The system will accept multiple delivery tasks. Each delivery will have a destination and a priority level.

Delivery	Destination	Priority
Package A	Location 5	Normal
Package B	Location 8	Urgent
Package C	Location 3	Normal
Package D	Location 10	High

2.3 Find the Best Route

For every delivery, the system will find an efficient path from the drone's current location to the destination.

Algorithms such as Dijkstra and A* can be used to calculate the best route. The system will not simply move the drone randomly; it will calculate the route using the graph and its edge weights.

```
Route 1 = 8 km
Route 2 = 6 km
Route 3 = 10 km
```

2.4 Consider Battery / Energy Usage

The shortest route may not always be the best route. Therefore, the system can consider both distance and energy usage while choosing a route.

Route A Distance = 5 km Energy Usage = 40% Shorter, but higher energy use.	Route B Distance = 6 km Energy Usage = 25% Slightly longer, but more energy-efficient.
--	--

2.5-2.7 Priority, Multi-Drone Assignment & Output

2.5 Prioritize Deliveries

Some packages may be more important than others. A Priority Queue and a Greedy approach can be used so that important deliveries are processed before normal deliveries.

1. Emergency Medicine
2. Express Delivery
3. Normal Package

2.6 Divide Work Between Multiple Drones

Instead of giving all work to one drone, the system can divide the tasks across available drones. The goal is to use all drones efficiently and complete the deliveries faster.

Available drones Drone 1 Drone 2 Drone 3	Possible assignment Drone 1 - Package A, Package D Drone 2 - Package B, Package F Drone 3 - Package C, Package E
--	--

2.7 Show the Selected Route and Delivery Information

After making a decision, the system can display the chosen drone and all key information. This makes the algorithm's decision easy to understand during the project demonstration.

Selected Drone: Drone 2
 Package: PKG-005
 Destination: Location 8
 Route: Warehouse -> A -> C -> F -> Customer
 Distance: 6.2 km
 Estimated Energy Usage: 24%
 Priority: High
 Algorithm Used: A*



3

Algorithms and Data Structures Used

3 Algorithms Used

Problem	Algorithm / Data Structure
Represent city	Graph (Adjacency List)
Find shortest route	Dijkstra's Algorithm
Faster route search	A* Algorithm
Select important delivery	Greedy Algorithm
Manage delivery priority	Priority Queue
Assign tasks to drones	Greedy Strategy
Store locations and paths	Graph / Adjacency List



Algorithms
power smarter
deliveries!

4 Main Benefits of the Project



Reduced delivery time – Find better routes instead of random travel.



Reduced battery usage – Energy-efficient routing increases the number of possible deliveries.



Better delivery priority – Urgent packages (e.g., medicine) are delivered first.



Better use of multiple drones – Tasks are distributed efficiently.



Reduced operational cost – Less distance and energy consumption lowers costs.



Automatic decision-making – No manual route planning or task assignment needed.



Real-world implementation of algorithms – Practical use of DAA concepts in a real-world scenario.

Problem	Algorithm / Data Structure	Purpose
Represent the city	Graph	Stores locations and connections between them
Store graph connections	Adjacency List	Efficiently stores which locations are connected
Find shortest route	Dijkstra's Algorithm	Finds the minimum-cost path in a weighted graph
Faster goal-directed route search	A* Algorithm	Uses a heuristic to guide the search toward the destination
Select important delivery	Greedy Algorithm	Makes a suitable immediate choice based on priority/cost
Manage delivery priority	Priority Queue	Keeps urgent deliveries ahead of normal deliveries
Assign tasks to drones	Greedy Strategy	Selects a suitable available drone for a delivery

How the Algorithms Work Together

Delivery Requests	->	Priority Queue	->	Select Delivery	->	Choose Drone
Check Energy	->	Dijkstra / A*	->	Find Best Route	->	Complete Delivery

The important point is that one algorithm is not doing everything. Different algorithms and data structures solve different parts of the delivery problem.

4

Main Benefits of the Project

4.1 Reduced Delivery Time

The system calculates better routes instead of allowing drones to travel through inefficient paths.

Benefit: Packages can reach customers faster.

4.2 Reduced Battery Usage

Drone battery is limited. The system can consider energy usage while selecting routes.

Benefit: A drone may be able to complete more deliveries before it needs to recharge.

4.3 Better Delivery Priority

Urgent deliveries, such as medicine or emergency supplies, can be processed before normal deliveries.

Benefit: Important packages are not delayed behind low-priority deliveries.

4.4 Better Use of Multiple Drones

The system distributes delivery tasks between available drones instead of overloading one drone.

Benefit: Multiple deliveries can happen in parallel and the drone fleet is used more efficiently.

4.5 Reduced Operational Cost

Reducing unnecessary travel distance and energy usage can reduce delivery costs.

Benefit: The same drone fleet can perform more work with fewer wasted resources.

4.6 Automatic Decision-Making

A human operator does not need to manually calculate every route and assign every package.

Benefit: The system automatically supports route selection, priority management, and drone assignment.

4.7 Real-World Implementation of Algorithms

Instead of demonstrating Dijkstra, A*, Greedy, Graphs, and Priority Queues as separate classroom examples, this project combines them inside one practical scenario.

Benefit: It clearly shows how Design and Analysis of Algorithms (DAA) concepts can solve a real-world logistics problem.

Complete Visual Summary

The full infographic below summarizes the complete project at a glance. It is useful for presentation, revision, and explaining the project quickly to your teacher.



1 Project in Simple Words

Our project is a **Drone Delivery Route Optimization System** that helps delivery drones find the most efficient routes to deliver packages in a city. The system automatically decides which delivery to do first, which route to take, and how to divide tasks among multiple drones while considering distance, travel time, and battery (energy) usage.

"It's like a smart planner for delivery drones."

Real-World Scenario



2 What the System Will Do



Create a Simulated City Map

Represent the city as a graph where locations are nodes and paths are edges with distance, time, and energy information.



Receive Delivery Requests

Take multiple delivery tasks with different destinations and priorities.



Find the Best Route

Use algorithms like Dijkstra or A* to find the shortest and fastest route.



Consider Battery/Energy Usage

Choose routes that are energy-efficient, not just the shortest distance.



Prioritize Deliveries

Use a greedy approach and priority queue to deliver urgent packages first (e.g., medicine, food).



Divide Work Between Multiple Drones

Assign deliveries to different drones efficiently so all packages are delivered on time.



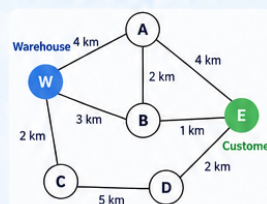
Show the Selected Route

Display the chosen route, drone details, distance, energy usage, and delivery status.

How It Works (Flow)



Example: Finding the Best Route



Possible Routes (W → E)

Route 1: W → A → E = 8 km
Route 2: W → B → E = 4 km ✓ (Shortest)
Route 3: W → C → D → E = 9 km

The system will choose **Route 2** because it is the shortest and most efficient.

Distance vs Energy Example

Route A

Distance: 5 km
Energy: 40%



Shorter, but uses more energy

Route B

Distance: 6 km
Energy: 25%



A bit longer, but more energy efficient

3 Algorithms Used

Problem	Algorithm / Data Structure
Represent city	Graph (Adjacency List)
Find shortest route	Dijkstra's Algorithm
Faster route search	A* Algorithm
Select important delivery	Greedy Algorithm
Manage delivery priority	Priority Queue
Assign tasks to drones	Greedy Strategy
Store locations and paths	Graph / Adjacency List



Algorithms power smarter deliveries!

4 Main Benefits of the Project

- Reduced delivery time** – Find better routes instead of random travel.
- Reduced battery usage** – Energy-efficient routing increases the number of possible deliveries.
- Better delivery priority** – Urgent packages (e.g., medicine) are delivered first.
- Better use of multiple drones** – Tasks are distributed efficiently.
- Reduced operational cost** – Less distance and energy consumption lowers costs.
- Automatic decision-making** – No manual route planning or task assignment needed.
- Real-world implementation of algorithms** – Practical use of DAA concepts in a real-world scenario.